

Lecture 16

Coordinate Bethe ansatz II: Lieb–Liniger and Yang–Baxter

Objectives. The method of [Lecture 15](#) works for particles on a line. Bosons with a contact interaction collide pairwise, each collision multiplies the wave by a phase, and the Bethe equations follow as before — this time with a proof that the wavefunction is an eigenfunction. Then comes the question the whole algebraic method rests on. If the particles carry a label, such as a spin, a collision is a matrix, and three-body scattering can factorise into two-body collisions only if the two orders of [Figure 5](#) give the same matrix. That condition is the Yang–Baxter equation, and here it is derived, not postulated.

16.1 The gas

N identical bosons move on a ring of length L and interact only when they touch:

$$H = - \sum_{j=1}^N \frac{\partial^2}{\partial x_j^2} + 2c \sum_{i<j} \delta(x_i - x_j), \quad c > 0. \quad (16.1)$$

This is the *Lieb–Liniger* model (1963), repulsive for $c > 0$, in units $\hbar = 2m = 1$: a free particle of momentum k has energy k^2 and velocity $2k$. **A named clash, the kinetic normalisation:** [Lecture 21](#) uses $\hbar = m = 1$, kinetic term $-\frac{1}{2} \sum \partial^2$, to match [Lecture 11](#).

It is the gas § 0.2 promised. In the thin tubes of the quantum Newton’s cradle ([Figure 2](#)) the transverse motion is frozen, and the atoms’ short-range interaction becomes, along the tube, a delta of a strength fixed by the scattering length and the tube width (Olshanii 1998) · *cited*. The lecture uses criterion (S) of § 13.5. Between collisions the particles are free; the delta imposes a condition wherever two of them meet.

16.2 Two bosons

The contact condition

For two particles use the centre of mass $X = \frac{1}{2}(x_1 + x_2)$ and the separation $r = x_2 - x_1$. Then

$$-\partial_1^2 - \partial_2^2 = -\frac{1}{2}\partial_X^2 - 2\partial_r^2, \quad H = -\frac{1}{2}\partial_X^2 - 2\partial_r^2 + 2c \delta(r). \quad (16.2)$$

Integrate $H\psi = E\psi$ over $-\epsilon < r < \epsilon$ and let $\epsilon \rightarrow 0$: $-2[\partial_r \psi]_{r=0^-}^{r=0^+} + 2c \psi|_{r=0} = 0$. (ψ itself must be continuous: a jump would put a δ' into $\partial_r^2 \psi$ that nothing cancels.) So the r -derivative jumps by $c\psi$. For bosons ψ is even in r , its derivative odd, and the jump is twice the one-sided value. With $\partial_r = \frac{1}{2}(\partial_2 - \partial_1)$ this is the *contact condition*

$$(\partial_2 - \partial_1 - c)\psi = 0 \quad \text{at } x_2 = x_1, \text{ approached from } x_1 < x_2. \quad (16.3)$$

A symmetric ψ is fixed by its values on $x_1 < x_2$, so the problem is: solve the free equation on $x_1 < x_2$, subject to (16.3) on its edge (Figure 16.1).

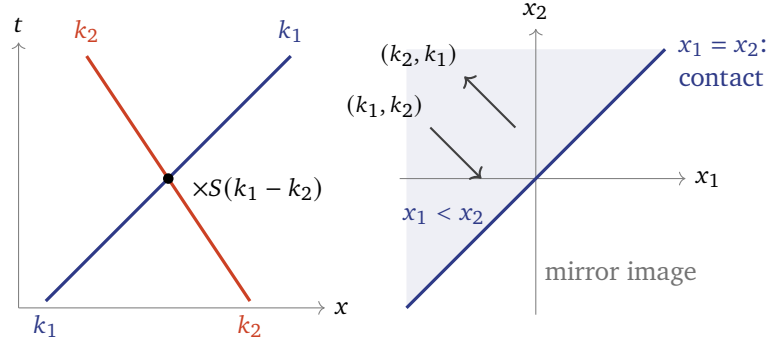


Figure 16.1: Two bosons. Left: the two momentum lines in space and time; the faster comes from the left, and after the collision the same two momenta travel on. All the collision can do is multiply the wave by $S(k_1 - k_2)$. Right: the configuration plane. The wavefunction is free on $x_1 < x_2$; on its edge the delta imposes (16.3), tying the wave (k_2, k_1) to the incoming (k_1, k_2) . The other half is the mirror image.

The two-body S-matrix

On $x_1 < x_2$ try, as in Lecture 15,

$$\psi(x_1, x_2) = A_{12} e^{i(k_1 x_1 + k_2 x_2)} + A_{21} e^{i(k_2 x_1 + k_1 x_2)}, \quad E = k_1^2 + k_2^2. \quad (16.4)$$

Both waves solve the free equation with the same energy; only the contact condition is left.

Proposition 16.1 (The two-body S-matrix). *The wavefunction (16.4) satisfies (16.3) if and only if*

$$\frac{A_{21}}{A_{12}} = S(k_1 - k_2), \quad S(k) = \frac{k - ic}{k + ic}. \quad (16.5)$$

Proof. At $x_1 = x_2 = x$ both waves are $e^{i(k_1 + k_2)x}$, and $\partial_2 - \partial_1$ brings down $i(k_2 - k_1)$ from the first and $i(k_1 - k_2)$ from the second. With $k = k_1 - k_2$ the condition is $A_{12}(-ik - c) + A_{21}(ik - c) = 0$, so $A_{21}/A_{12} = (ik + c)/(ik - c) = (k - ic)/(k + ic)$. ■

S is Proposition 15.1's formula at $\lambda = k/c$. No rapidity is needed: by Galilean invariance, which the lattice lacks, the phase depends on the momenta only through their difference. Its properties:

- $|S(k)| = 1$ for real k , and $S(k)S(-k) = 1$. Write

$$S(k) = -e^{i\theta(k)}, \quad \theta(k) = 2 \arctan \frac{k}{c}, \quad \theta'(k) = \frac{2c}{k^2 + c^2} > 0 \quad (16.6)$$

(Figure 16.2, left).

- $S(0) = -1$: with equal momenta $A_{21} = -A_{12}$ and the wave vanishes — a Pauli principle for bosons, which nobody put in.
- As $c \rightarrow \infty$, $S \rightarrow -1$: the wavefunction vanishes where two particles meet, and these *hard-core bosons* have the energies of free fermions (Girardeau 1960), the *Tonks–Girardeau gas*. As $c \rightarrow 0$, $S \rightarrow 1$: free bosons.

Reading

Reflection and transmission. Were the two particles distinguishable, with the same interaction, the relative wave coming in from $r > 0$ would be partly transmitted to $r < 0$ and partly reflected, with amplitudes (the same integration)

$$t(k) = \frac{k}{k + ic}, \quad r(k) = \frac{-ic}{k + ic}, \quad |t|^2 + |r|^2 = 1.$$

For identical bosons a transmitted pair and a reflected pair are the same state, both amplitudes feed the one outgoing wave, and $t + r = S$.

What the phase does: a displacement

A plane wave has no position, so superpose: a narrow spread of k_1 around $\bar{k}_1$ and of k_2 around $\bar{k}_2 < \bar{k}_1$, the faster particle on the left. A packet sits where its phase is stationary in the momentum. In the incoming term the k_1 phase is $k_1 x_1 - k_1^2 t$, stationary at $x_1 = 2\bar{k}_1 t$. In the outgoing term it is $k_1 x_2 - k_1^2 t + \arg S(k_1 - k_2)$, and $\arg S = \pi + \theta$, so after the collision

• *sketched*

$$x_2 = 2\bar{k}_1 t - \theta'(\bar{k}_1 - \bar{k}_2), \quad x_1 = 2\bar{k}_2 t + \theta'(\bar{k}_1 - \bar{k}_2) \quad (16.7)$$

Each momentum comes out on a line parallel to its incoming one: the line carrying the larger momentum is shifted by $-\theta'$, the other by $+\theta'$ — in the centre-of-mass frame, both particles are delayed (Figure 16.2, right, checks this against numerically superposed exact solutions).

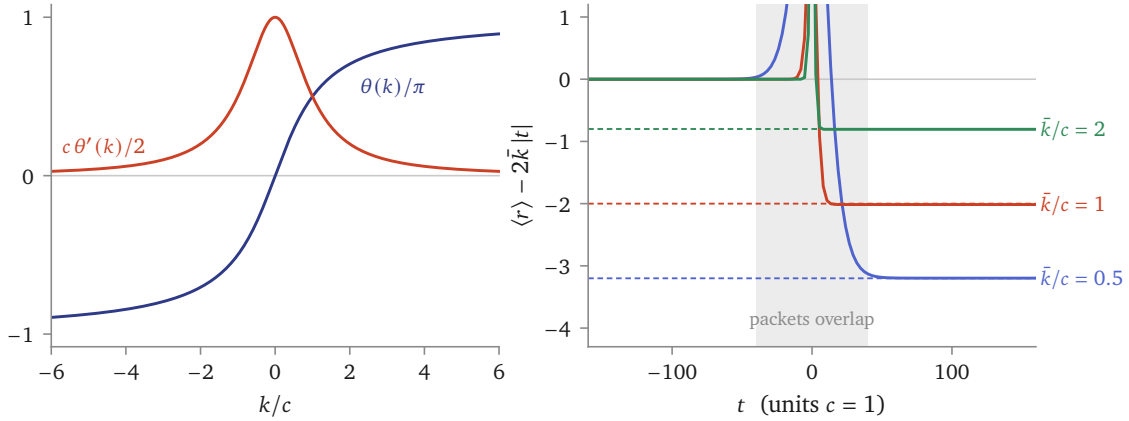


Figure 16.2: Left: the phase $\theta(k) = 2 \arctan(k/c)$, $S = -e^{i\theta}$, and its derivative. Right: two bosons, centre-of-mass frame, a Gaussian packet in the relative momentum, the exact two-body solution superposed numerically on $r = x_2 - x_1 > 0$. The centre $\langle r \rangle$ minus its hard-core value $2\bar{k}|t|$, $\bar{k} = \bar{k}_1 - \bar{k}_2$, starts at 0 and ends on $-\theta'(\bar{k})$ (dashed), the two displacements (16.7); shaded, the packets overlap. (scripts/figures/lieb_liniger.py asserts both asymptotes.)

This is the quantum form of Proposition 9.8, where the Toda pair exchanged its momenta and the fast line came out displaced by $-2 \ln \nu$: in one dimension a two-body collision only hands over the momenta and shifts the lines. For the delta gas this is an analogy — classical point particles with a contact interaction show no displacement at all (Sutherland §4.2). For the Toda pair itself the reading at the end shows $\theta' \rightarrow 2 \ln \nu$ as $\hbar \rightarrow 0$.

16.3 N bosons

The fundamental domain

Cut the ring at an origin, so that $0 \leq x_j < L$. The $\binom{N}{2}$ planes $x_i = x_j$ cut this cube into $N!$ sectors, one for each order of the particles; periodicity glues cyclically related sectors, which is where the Bethe equations will come from. A symmetric ψ is fixed by its values on the *fundamental domain* $x_1 < \dots < x_N$, whose images under permutations are the other sectors (Figure 16.3); it has $N - 1$ walls, $x_j = x_{j+1}$.

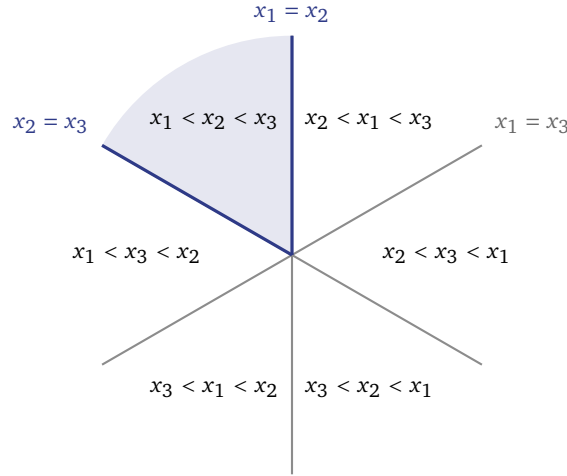


Figure 16.3: Three particles, seen along $(1, 1, 1)$: the planes $x_i = x_j$ cut the plane of relative coordinates into six sectors. The fundamental domain $x_1 < x_2 < x_3$ (shaded) has two walls; the other sectors are its images, and a symmetric wavefunction is fixed by its values there.

Definition 16.2 (Eigenfunctions of the delta gas). A symmetric, continuous function ψ , periodic with period L in every coordinate and smooth up to the walls within each sector, is an *eigenfunction* of (16.1) with eigenvalue E if $-\sum_j \partial_j^2 \psi = E\psi$ inside every sector, and on every wall $x_i = x_j$, approached from the side $x_i < x_j$, $(\partial_j - \partial_i - c)\psi = 0$.

The wall condition is § 16.2's, wall by wall. Rigorously H is defined as a self-adjoint operator through its quadratic form, in which the places where three particles meet have measure zero and impose nothing (the KLMN theorem; Reed and Simon, vol. II, §X.2) · cited.

The Bethe wavefunction

On the fundamental domain Bethe's ansatz is Equation (15.14) with momenta for rapidities:

$$\psi(x_1, \dots, x_N) = \sum_{\sigma \in S_N} A_\sigma \exp\left(i \sum_{j=1}^N k_{\sigma(j)} x_j\right), \quad A_\sigma = \prod_{\substack{a < b \\ \sigma \text{ puts } b \text{ before } a}} S(k_a - k_b), \quad (16.8)$$

one factor S for each pair of momenta that σ has exchanged.

Theorem 16.3 (Lieb and Liniger 1963). Let $k_1, \dots, k_N$ be pairwise distinct, with no two

differing by $\pm ic$, and let them satisfy the Bethe equations

$$e^{ik_j L} = \prod_{l \neq j} \frac{k_j - k_l + ic}{k_j - k_l - ic}, \quad j = 1, \dots, N. \quad (16.9)$$

Then the symmetric extension of (16.8), if it is not zero, is an eigenfunction of the delta gas on the ring in the sense of Definition 16.2, with

$$E = \sum_{j=1}^N k_j^2, \quad K = \sum_{j=1}^N k_j. \quad (16.10)$$

Proof. Inside the sectors every term is a plane wave of energy $\sum k_j^2$. On the wall $x_{j+1} = x_j$ pair each σ with $\sigma' = \sigma \circ (j \ j+1)$, which exchanges k_a , $a = \sigma(j)$, and k_b , $b = \sigma(j+1)$. On the wall the two terms share an exponential, which no other term has, and together they are a two-body wave in (x_j, x_{j+1}) times a factor free of x_j, x_{j+1} . By Proposition 16.1 the contact condition holds if and only if $A_{\sigma'}/A_{\sigma} = S(k_a - k_b)$ for every σ ; the product in (16.8) gains or loses exactly the factor of the pair $\{a, b\}$, which is $S(k_a - k_b)$ either way since $S(k)S(-k) = 1$. The other walls are images of these, and the condition of Definition 16.2 is symmetric, so it goes with them. Periodicity, $\psi(x_1, \dots, x_N) = \psi(x_2, \dots, x_N, x_1 + L)$, is the periodicity step of Theorem 15.3 verbatim: moving k_a from last place to first passes it once past every other momentum, giving $e^{ik_a L} = \prod_{b \neq a} S(k_b - k_a)$, which is (16.9). ■

Unlike Theorem 15.3, this proof is complete: each wall is treated alone, since Definition 16.2 does not see where walls meet, while on the lattice three adjacent spins were a real configuration (§ 15.4). The check script verifies the walls for $N = 3, 4$ symbolically; the tutorial compares $N = 2$ with a computer.

All roots are real, and each set of quantum numbers gives one state

Lecture 15 had strings, complex roots making bound states. Repulsion forbids them.

Proposition 16.4 (Real roots). For $c > 0$, every solution of (16.9) is real.

Proof. Let k_j have the largest imaginary part, m , and suppose $m > 0$. The left side of (16.9) has modulus $e^{-mL} < 1$. On the right, $\text{Im}(k_j - k_l) = d \geq 0$ for every l , and $|x + i(d+c)| \geq |x + i(d-c)|$ for real x because $d, c \geq 0$; so every factor has modulus at least 1, a contradiction. Hence $m \leq 0$. The same argument at the smallest imaginary part, with every inequality reversed, gives the smallest imaginary part ≥ 0 . So all are 0 (Korepin, Bogoliubov and Izergin, §I.2). ■

For attractive $c < 0$ the inequalities reverse and the proof fails: complex roots appear, in strings, and they are bound states of bosons (McGuire 1964). That gas is not treated here.

With real roots take logarithms, using $(k + ic)/(k - ic) = -e^{-i\theta(k)}$:

$$k_j L + \sum_{l=1}^N \theta(k_j - k_l) = 2\pi I_j, \quad I_j \in \mathbb{Z} + \frac{N-1}{2}, \quad (16.11)$$

the $N - 1$ signs going into I_j ($\theta(0) = 0$, so the term $l = j$ is harmless). Summing over j , the θ 's cancel in pairs and $K = (2\pi/L) \sum_j I_j$.

Proposition 16.5 (Yang and Yang 1969). For $c > 0$ and any real $I_1, \dots, I_N$, (16.11) has exactly one solution. If $I_j > I_l$ then $k_j > k_l$; in particular distinct quantum numbers give distinct roots, and equal quantum numbers give equal roots.

Proof (★). Let $\Theta(x) = \int_0^x \theta$, an even function growing at most linearly. Then (16.11) says $\partial B / \partial k_j = 0$ for

$$B(k) = \frac{L}{2} \sum_j k_j^2 - 2\pi \sum_j I_j k_j + \frac{1}{2} \sum_{j,l} \Theta(k_j - k_l),$$

whose Hessian is $L \mathbb{1} + \sum_{j < l} \theta'(k_j - k_l) (e_j - e_l)(e_j - e_l)^T$: L times the identity plus a sum of positive semidefinite matrices, since $\theta' > 0$. So B is strictly convex and grows like $\frac{L}{2} |k|^2$; it has exactly one critical point, its minimum. For the order, subtract the equations for j and l : $(k_j - k_l)L + \sum_m [\theta(k_j - k_m) - \theta(k_l - k_m)] = 2\pi(I_j - I_l)$. Since θ increases, the left side has the sign of $k_j - k_l$. ■

The reason: (16.11) is the gradient of a strictly convex function of $k_1, \dots, k_N$, which has exactly one minimum.

So for $c > 0$ the Bethe states are labelled exactly by sets of N distinct $I_j \in \mathbb{Z} + \frac{N-1}{2}$, one real solution each; coincident roots come only from coincident I 's, and then $S(0) = -1$ kills the vector. These are the quantum numbers of N free fermions, the $c \rightarrow \infty$ limit $k_j = 2\pi I_j / L$. That the Bethe states are *all* the eigenstates is here a theorem (Dorlas 1993, who also proves them orthogonal) · *cited*, unlike for the Heisenberg chain (§ 15.6). The ground state is the most compact set, $I_j = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$, at every $c > 0$ (Yang and Yang 1969) · *cited*; Figure 16.4 follows its roots.

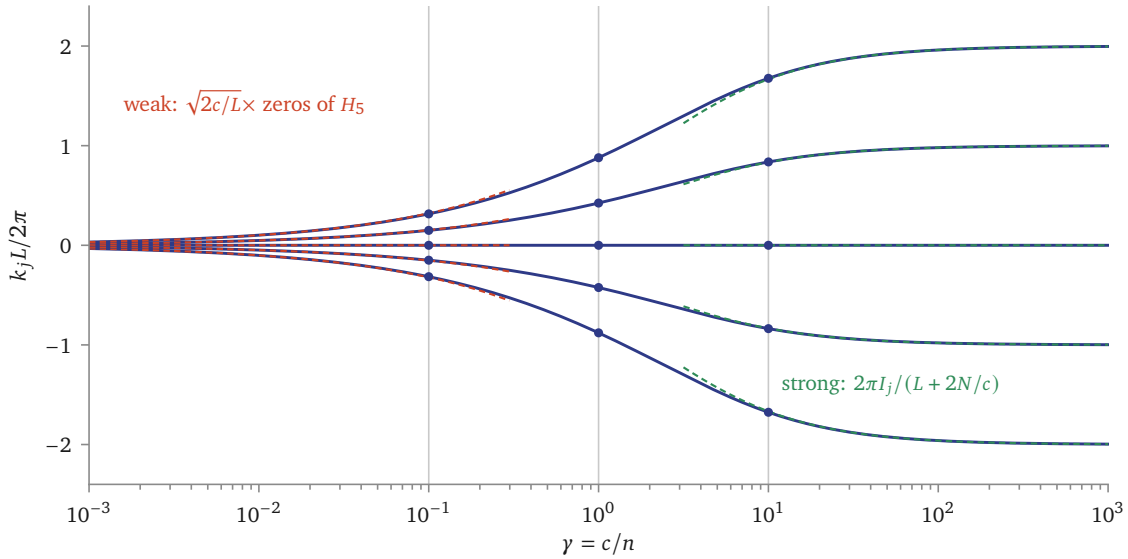


Figure 16.4: Ground-state roots of $N = 5$ bosons, $L = 5$ (density $n = 1$), against $\gamma = c/n$; dots at $\gamma = 0.1, 1, 10$. Strong coupling: towards the free-fermion $2\pi I_j / L$, as $2\pi I_j / (L + 2N/c)$ (green). Weak coupling: $\sqrt{2c/L}$ times the zeros of the Hermite polynomial H_5 (red); the tutorial derives both. (scripts/figures/lieb_liniger.py asserts the equations and both limits; random starts agree at the three marked couplings.)

Rapidities. On a ring the particles never become free, yet every eigenstate carries a set $\{k_j\}$, the *rapidities*, which enter E and K as if they were: the quantum counterpart of the

asymptotic momenta of [Theorem 9.9](#). They are conserved in the sense that matters: the Bethe states form an orthogonal basis (Dorlas 1993) · *cited*, so in any state the probability of each set $\{k_j\}$ is the weight of one eigenstate, constant in time. That conserved distribution is what § 0.2 said would stop the cradle’s gas relaxing; [Lecture 23](#) builds on it.

16.4 Why one dimension

§ 0.2 counted. For two equal masses, momentum and energy are two equations for two outgoing momenta, solved only by the incoming pair or its exchange ([Proposition 0.1](#)). Quantum mechanically the same holds for *any* short-range pair interaction, once the particles are free again: the outgoing wave of two particles contains only the momenta k_1, k_2 , and scattering fixes nothing but amplitudes.

For three particles the count fails: two equations for three unknowns leave a circle of outgoing momenta, the six rearrangements being six points on it. A generic three-body collision sends waves out over the whole circle — *diffraction* — which is how a one-dimensional gas can thermalise (Sutherland, §2.2). **Non-diffraction is not free.** It is what the Bethe wavefunction asserts — in (16.8) only rearrangements of the incoming momenta ever appear — and [Theorem 16.3](#) proves that the delta gas has it. One caution: the two-body count uses $E = k^2$; on a lattice even two magnons are not forced by kinematics (§ 15.3, at $K = \pi$).

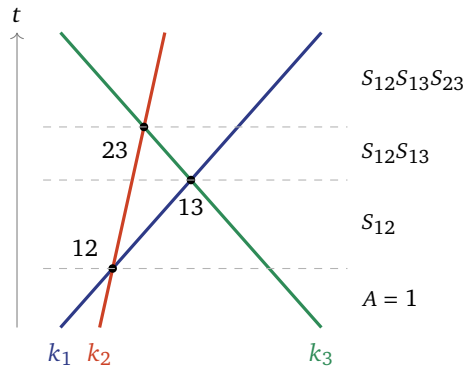


Figure 16.5: The Bethe wavefunction as a scattering event. Three momentum lines, $k_1 > k_2 > k_3$, cross pairwise; $S_{ab} = S(k_a - k_b)$. For a superposition of separated wave packets, between collisions the packet is carried by one term of (16.8), with amplitude A_σ (right): one factor per crossing passed. No other momenta appear. It is [Figure 9.6](#) with a phase where the Toda chain had a displacement; the other order of crossings gives the same product, because numbers commute.

In [Figure 16.5](#) each crossing multiplies the amplitude by one S . Had the lines crossed in the other order — the other panel of [Figure 5](#) — the same factors would come in another order, and nothing would change: for bosons without labels three-body consistency is automatic. It stops being automatic when a collision can do more than multiply by a number.

16.5 Particles with a label

Let each particle carry a label in $\mathbb{C}^d$ — a spin component, a hyperfine state — and let the delta be blind to it. A two-particle wavefunction takes values in $\mathbb{C}^d \otimes \mathbb{C}^d$, the first factor for

particle 1's label, and is symmetric (bosons) or antisymmetric (fermions) under exchanging positions and labels together:

$$\psi(x_1, x_2) = \pm P \psi(x_2, x_1), \quad (16.12)$$

with P the permutation of the two factors (Definition 8.3). On $x_1 < x_2$ write

$$\psi(x_1, x_2) = A e^{i(k_1 x_1 + k_2 x_2)} + A' e^{i(k_2 x_1 + k_1 x_2)}, \quad A, A' \in \mathbb{C}^d \otimes \mathbb{C}^d,$$

and let (16.12) fix ψ on $x_2 < x_1$.

Proposition 16.6 (Yang's matrix). *The delta interaction forces $A' = Y(k_1 - k_2)A$, with*

$$Y_B(k) = \frac{kP - ic}{k + ic} \text{ (bosons)}, \quad Y_F(k) = -\frac{kP + ic}{k + ic} \text{ (fermions)}. \quad (16.13)$$

Proof. P commutes with everything here, so split A, A' by $\Pi_{\pm} = \frac{1}{2}(1 \pm P)$ into parts symmetric and antisymmetric in the labels. Where (16.12) makes ψ symmetric in the positions — Π_+ for bosons, Π_- for fermions — the problem is Proposition 16.1's, and $A' = S(k)A$. In the other part ψ is antisymmetric in the positions and vanishes at $x_1 = x_2$, so the delta does not act, and continuity forces $A' = -A$. Hence $Y_B = S\Pi_+ - \Pi_-$ and $Y_F = S\Pi_- - \Pi_+$. With $S - 1 = -2ic/(k + ic)$ and $S + 1 = 2k/(k + ic)$ these are (16.13). ■

For spin- $\frac{1}{2}$ fermions, then, the interacting phase lives in the spin singlet, and the triplet scatters as free fermions do: two fermions with parallel spins never meet.

Reading

In terms of the t and r of § 16.2's reading, $Y_B = t(k)P + r(k)$: transmission carries the labels across the collision, so the particle now on the left has the label that was on the right — the P — and reflection leaves them where they were. For fermions $Y_F = -t(k)P + r(k)$.

Following the momentum. Y is written in the order of positions (slot 1 is the particle on the left). Figure 16.5 follows the momenta, and there the natural object is

$$X(u) = PY, \quad u = \frac{k_1 - k_2}{c} : \quad X_B(u) = \frac{R(u)}{u + i}, \quad X_F(u) = \frac{R(-u)}{u + i} = -\frac{L(u)}{u + i}, \quad (16.14)$$

where $R(u) = u - iP$ is the course's rational R-matrix and $L(u) = u + iP$ the Lax operator of the Heisenberg chain (Lecture 17). Both obey unitarity, $X(u)X(-u) = 1$. X_B is the R-matrix of Lecture 17; X_F , which Yang found for spin- $\frac{1}{2}$ fermions and which is Lecture 0's $u + iP$ up to a scalar, is why the algebraic Bethe ansatz of Lecture 18 solves the gases of Lecture 22.

16.6 Three-body consistency: the Yang–Baxter equation

Three labelled particles, on $x_1 < x_2 < x_3$, with the factorised ansatz

$$\psi = \sum_{\sigma \in S_3} A_{\sigma} e^{i(k_{\sigma(1)}x_1 + k_{\sigma(2)}x_2 + k_{\sigma(3)}x_3)}, \quad A_{\sigma} \in (\mathbb{C}^d)^{\otimes 3}, \quad k_1 > k_2 > k_3. \quad (16.15)$$

The term $\sigma = \text{id}$, fastest on the left, is the *incoming* wave; $\sigma = w_0$, the order reversed, is the *outgoing* one; the other four come after one or two collisions. The ansatz is non-diffraction written down; the question is whether the contact conditions allow it.

The walls. On the wall $x_{j+1} = x_j$ the terms pair up as before, σ with σs_j , $s_j = (j \ j+1)$, and the third particle is a spectator. So [Proposition 16.6](#), applied to the labels in slots $j, j+1$, says

$$A_{\sigma s_j} = Y_{j,j+1}(k_{\sigma(j)} - k_{\sigma(j+1)})A_{\sigma} \quad (16.16)$$

for every σ and j : all the conditions. Each is invertible, and read from σs_j it says the same thing: it is one contact condition.

Following the momenta. Reorder the slots of A_{σ} so that slot a holds the label of the particle that carries k_a : $B_{\sigma} = \Pi_{\sigma} A_{\sigma}$, where Π_{σ} moves the factor in slot j to slot $\sigma(j)$.

Lemma 16.7 (A crossing is an X). In terms of B , (16.16) reads

$$B_{\sigma s_j} = X_{ab}(u_{ab}) B_{\sigma}, \quad a = \sigma(j), \quad b = \sigma(j+1), \quad u_{ab} = \frac{k_a - k_b}{c},$$

with X_{ab} acting on slots a and b (first leg on a) ([Lecture 8's](#) leg notation).

Proof. $\Pi_{\sigma s_j} = \Pi_{\sigma} P_{j,j+1}$: both send slot j to slot $\sigma(j+1)$ and slot $j+1$ to $\sigma(j)$. So $B_{\sigma s_j} = \Pi_{\sigma} P_{j,j+1} Y_{j,j+1} A_{\sigma} = \Pi_{\sigma} X_{j,j+1} \Pi_{\sigma}^{-1} B_{\sigma}$, and conjugating by Π_{σ} moves the legs of X from slots $j, j+1$ to $\sigma(j), \sigma(j+1)$. ■

So each crossing of momentum lines a and b multiplies by $X_{ab}(u_{ab})$: [Figure 16.5](#) with matrices. From the incoming wave to the outgoing one there are two routes. Along $\text{id} \rightarrow s_1 \rightarrow s_1 s_2 \rightarrow w_0$ the crossings are 12, 13, 23; along $\text{id} \rightarrow s_2 \rightarrow s_2 s_1 \rightarrow w_0$ they are 23, 13, 12 ([Figure 16.6](#)): the two panels of [Figure 5](#).

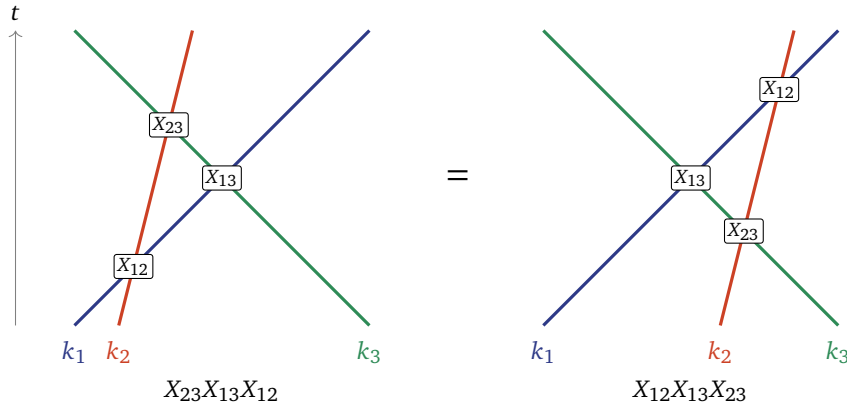


Figure 16.6: The Yang–Baxter equation, read off three labelled particles. Each line carries a momentum and a label; a crossing of lines a and b acts on their labels with $X_{ab}(u_{ab})$, $u_{ab} = (k_a - k_b)/c$ ([Lemma 16.7](#)). Time runs up, so the first crossing is the rightmost factor. The two orders are the two routes from the incoming wave to the outgoing one; the contact conditions allow a wavefunction only if they agree.

Theorem 16.8 (Three-body consistency). Let the two-body problem give a relation of the form (16.16) with $X = PY$ invertible and unitary, $X_{21}(-u)X_{12}(u) = \mathbb{1}$ — which for X commuting with P , as Yang’s do, is $X(u)X(-u) = \mathbb{1}$. The contact conditions on the three-body ansatz (16.15) have a solution for every incoming amplitude A_{id} if and only if

$$X_{12}(u_{12}) X_{13}(u_{13}) X_{23}(u_{23}) = X_{23}(u_{23}) X_{13}(u_{13}) X_{12}(u_{12}). \quad (16.17)$$

The outgoing amplitude is then $B_{w_0} = X_{12}X_{13}X_{23} B_{\text{id}}$.

Proof. The six amplitudes sit at the vertices of a hexagon, the permutations of S_3 , and the conditions sit on its six edges, each an invertible map from one vertex to the next (Lemma 16.7); unitarity says that the map read from the other end is its inverse, so each edge carries one relation. Given B_{id} , going round either side fixes everything up to B_{w_0} ; the two values agree for every B_{id} if and only if the two products of Figure 16.6 are equal, which is (16.17). If they are equal, the amplitudes so defined satisfy every edge, and $A_\sigma = \Pi_\sigma^{-1} B_\sigma$. ■

This is “derived, not postulated”: nothing was assumed but that the three-body wave is a sum of rearranged plane waves, and the equation is what that costs. The logic runs one way: the equation is *necessary* for a consistent factorised ansatz; it does not by itself prove that an interaction is non-diffractive.

Proposition 16.9 (Yang’s matrix satisfies it). X_B and X_F of (16.14) satisfy (16.17) for all real k_1, k_2, k_3 , for every d .

Proof. The scalar factors $1/(u_{ab} + i)$ are the same three on both sides. For X_B what is left is (16.17) for $R(u) = u - iP$. Expand both sides; the transpositions obey $P_{12}P_{13} = P_{13}P_{23} = P_{23}P_{12} =: C$ and $P_{13}P_{12} = P_{12}P_{23} = P_{23}P_{13} = C^{-1}$, and $P_{12}P_{13}P_{23} = P_{13} = P_{23}P_{13}P_{12}$. The terms with no P , one P and three P ’s agree term by term. The terms with two are $-(u_{23} + u_{12})C - u_{13}C^{-1}$ on the left and $-u_{13}C - (u_{12} + u_{23})C^{-1}$ on the right, and they agree because

$$u_{13} = u_{12} + u_{23} :$$

the spectral parameters are differences of momenta, so they add. For X_F , replace every u by $-u$. ■

The proof used only the form $u - iP$ and the additivity of u ; the variant $u - i(1 + \epsilon u)P$, just as harmless-looking, fails (16.17) for every $\epsilon \neq 0$ (the check script; Lecture 0’s reading).

Scalar S : automatic. Without labels, $d = 1$, $P = 1$ and $X = (u - i)/(u + i) = S$: both sides are $S_{12}S_{13}S_{23}$ — Lecture 0’s remark as a line of algebra. The Toda chain is in the same position: no labels, numbers (displacements) at the crossings, and its braid relation (Figure 9.6) is the scalar, automatic case. The equation earns its keep only with labels.

N particles · sketched. The amplitudes sit on the permutations of S_N , the conditions on the edges $\sigma \rightarrow \sigma s_j$, and they are consistent if every closed walk multiplies to $\mathbb{1}$. By Coxeter’s presentation of S_N (cited from Gaudin §10.1) the closed walks are generated by an edge and back (unitarity, $X_{21}(-u)X_{12}(u) = \mathbb{1}$), the squares $s_j s_l = s_l s_j$, $|j - l| \geq 2$ (automatic: four different lines), and the hexagons $s_j s_{j+1} s_j = s_{j+1} s_j s_{j+1}$ (the Yang–Baxter equation). So the Yang–Baxter equation and unitarity make the N -body factorised ansatz consistent; the check script confirms it for $N = 4$.

16.7 Factorised scattering

The N -body S -matrix is the ordered product of two-body matrices, one per pair of momentum lines, in the order the lines cross,

$$S_N = \prod_{\text{crossings, in time order}} X_{ab}(u_{ab}), \quad (16.18)$$

and by the Yang–Baxter equation every order of crossings gives the same product: the many-body problem is determined by the two-body one. On a ring, carrying one particle round gives, as in [Theorem 16.3](#), a condition — now on a vector of labels: an eigenvalue problem for a product of X 's, a transfer matrix, solved by the nested algebraic Bethe ansatz ([Lecture 18](#), [Lecture 22](#)).

	Heisenberg chain (Lecture 15)	delta gas (this lecture)
particles	magnons on a lattice	bosons on a line
variable	rapidity λ , $e^{ik} = (\lambda + i/2)/(\lambda - i/2)$	momentum k
two-body S	$(\lambda - i)/(\lambda + i)$	$(k - ic)/(k + ic)$, the same at $k = c\lambda$
Bethe equations	$e^{ik_j N} \prod_{l \neq j} S(\lambda_j - \lambda_l) = 1$	$e^{ik_j L} \prod_{l \neq j} S(k_j - k_l) = 1$
energy	$-\sum 2J/(4\lambda_j^2 + 1)$	$\sum k_j^2$
complex roots	strings: bound states	none for $c > 0$
eigenfunction	sketched	proved
completeness	numerics; a reformulation (cited)	proved (cited)

Table 16.1: The two coordinate Bethe ansätze side by side.

Dictionary

The open Toda chain scatters with its momenta permuted, and the three-body shifts do not depend on the order of collisions ([E6](#), [Theorem 9.9](#), [Figure 9.6](#)) $\rightarrow$ factorised scattering ([16.18](#)) and the Yang–Baxter equation ([16.17](#)) (*type A, structural analogy: the Toda chain has no labels, so its braid relation is the scalar, automatic case of [Figure 16.5](#); the classical CYBE of [Lecture 8](#) is a condition on r , not this picture, and its row belongs to [Lecture 17](#)*)

The displacement $\mp 2 \ln v$ of the momentum lines in a two-body collision ([Proposition 9.8](#)) $\rightarrow$ the displacement $\mp \theta'(k)$ of a wave packet ([16.7](#)), where $S = -e^{i\theta}$ (*type A, sketched: for the delta gas an analogy; for the Toda pair itself the reading shows $\theta' \rightarrow 2 \ln v$ as $\hbar \rightarrow 0$*)

Tutorial

16.8 Solving the equations

Two particles by hand. Put $k_{1,2} = K/2 \pm q$ with $K = 2\pi n/L$. Then $e^{iKL/2} = (-1)^n$, and the first equation of ([16.9](#)) becomes $qL + \theta(2q) = \pi(n+1) \pmod{2\pi}$, one equation for $q > 0$ ($q = 0$ gives the zero vector). Its left side increases from 0, so each allowed right side gives one root, and $E = K^2/2 + 2q^2$. For $K = 0$ the ground state has $qL + 2 \arctan(2q/c) = \pi$: $q \rightarrow \pi/L$ as $c \rightarrow \infty$ (free fermions, $k = \pm\pi/L$) and $q \approx \sqrt{c/L}$ as $c \rightarrow 0$. The check script diagonalises two bosons on a ring of M sites, $-\Delta/a^2 + (2c/a)\delta$, with $a = L/M$, at $M = 200, 400, 800$, extrapolates in a , and finds the twelve lowest levels to within 6×10^{-9} of these.

Numerically, use damped Newton on ([16.11](#)): its Jacobian is the Hessian of the convex function B of [Proposition 16.5](#), positive definite, so every step is well defined; halve the step until the residual decreases. Start from the free-fermion values $2\pi I_j/L$.

Strong coupling. For $|k| \ll c$, $\theta(k) \approx 2k/c$, and with $\sum_l k_l = 0$ for the symmetric ground state ([16.11](#)) becomes $k_j(L + 2N/c) = 2\pi I_j$, with error $O(c^{-3})$: the gas behaves as free fermions on a ring lengthened by $2N/c$, each particle excluding a length $2/c$.

Weak coupling. For $cL \ll 1$ the roots of the ground state crowd together, but their differences, of order $\sqrt{c/L}$, stay large compared with c ; then $\theta(x) \approx \pi \operatorname{sgn} x - 2c/x$. The $\pi \operatorname{sgn}$ terms add up to exactly $2\pi I_j$ for the symmetric I 's and cancel, leaving $k_j L = \sum_{l \neq j} 2c/(k_j - k_l)$.

With $k_j = \sqrt{2c/L} h_j$ this is $h_j = \sum_{l \neq j} 1/(h_j - h_l)$, Stieltjes' characterisation of the zeros of the Hermite polynomial H_N (Gaudin §4.3). Figure 16.4 shows both limits.

Reading

16.9 The Zamolodchikov–Faddeev algebra

Factorised scattering has an algebraic shorthand. Give each asymptotic particle an operator $Z(k)$ that creates it, with one component per label, and write $Z_a(k)$ for $Z(k)$ with its label in tensor slot a (Lecture 8's leg notation). Postulate the exchange relation

$$Z_1(k_1) Z_2(k_2) = X_{12}(u_{12}) Z_2(k_2) Z_1(k_1) :$$

reordering two neighbours costs one two-body matrix. A product of three can be reversed in two ways, giving $X_{12}X_{13}X_{23} Z_3Z_2Z_1$ and $X_{23}X_{13}X_{12} Z_3Z_2Z_1$: the two sides of (16.17). Exchanging twice returns the product to itself only if $X(u)X(-u) = \mathbb{1}$ (for matrices that commute with P , as Yang's do): unitarity. This is the Zamolodchikov–Faddeev algebra (Zamolodchikov and Zamolodchikov 1979; Faddeev 1980) · *cited*. In relativistic field theory, where it was born, the two-body matrix obeys a third condition, *crossing symmetry*, which relates a particle going forward in time to an antiparticle going backwards. It has no counterpart for the non-relativistic particles of this course, which is one reason field theory is outside it (Lecture 0).

Reading

16.10 The Toda pair, quantised

The Dictionary's second row pairs θ' with the Toda displacement. For the Toda pair this can be checked. In Proposition 9.8 the relative coordinate $x = q_1 - q_2$ and its momentum $p = \frac{1}{2}(p_1 - p_2)$ have $H_{\text{rel}} = p^2 + e^x$, and the relative speed is $v = 2p$. Quantised,

$$-\hbar^2 \psi'' + e^x \psi = p^2 \psi.$$

With $\hbar = 1$ and $k = p$, the solution that decays under the wall is $\psi = K_{2ik}(2e^{x/2})$, the modified Bessel function: the chain rule turns this equation into Bessel's. Far from the wall, $x \rightarrow -\infty$, the small-argument form of K_ν gives · *cited*: DLMF §§10.25–10.27

$$\psi \approx \frac{1}{2} [\Gamma(-2ik) e^{ikx} + \Gamma(2ik) e^{-ikx}] :$$

an incoming wave e^{ikx} , reflected with the phase $\Gamma(2ik)/\Gamma(-2ik)$. Since $\Gamma(-2ik) = \overline{\Gamma(2ik)}$, the stationary-phase argument of § 16.2 puts the incoming packet on $x = 2kt + x_w$ and the outgoing one on $x = -2kt + x_w$, with $x_w = 2 \operatorname{Re} \psi_0(2ik)$ and $\psi_0 = \Gamma'/\Gamma$ the digamma function: the packet moves as if reflected by a hard wall at x_w . In the language of § 16.2, with $k = p_1 - p_2 = 2p$ the momentum difference, the reflected wave's phase is $\theta(k) = 2 \arg \Gamma(ik)$ up to a constant, and $\theta'(k) = 2 \operatorname{Re} \psi_0(ik) = x_w$: the wall is θ' . Restoring $\hbar$ shifts x by $2 \ln \hbar$ and replaces k by $p/\hbar$, so

$$x_w = 2 \ln \hbar + 2 \operatorname{Re} \psi_0(2ip/\hbar) \longrightarrow 2 \ln(2p) = 2 \ln v \quad (\hbar \rightarrow 0),$$

by Stirling’s formula, $\operatorname{Re} \psi_0(iy) = \ln y + O(y^{-2})$. The classical orbit of [Proposition 9.8](#), $x = \ln v^2 - v|t| + o(1)$, has its effective wall exactly there, and that wall is what displaces each momentum line by $\mp 2 \ln v$. So the classical displacement is the $\hbar \rightarrow 0$ limit of the quantum one (the check script verifies each step numerically; Sutherland §4.2 does the same for the whole hyperbolic family) · *sketched*.

Summary.

- *The gas*: $H = -\sum \partial_j^2 + 2c \sum \delta(x_i - x_j)$, $c > 0$, $\hbar = 2m = 1$. The delta is a contact condition, $(\partial_2 - \partial_1 - c)\psi = 0$ on the diagonal ([16.3](#)).
- *Two bosons*: $A_{21}/A_{12} = S(k_1 - k_2)$, $S(k) = (k - ic)/(k + ic) = -e^{i\theta(k)}$ ([Proposition 16.1](#)); a wave packet comes out displaced by $\mp \theta'(k)$, the counterpart of the Toda displacement (an analogy; for the Toda pair itself the reading gives the $\hbar \rightarrow 0$ limit).
- *N bosons*: the Bethe wavefunction is an eigenfunction under $e^{ik_j L} = \prod_{l \neq j} (k_j - k_l + ic)/(k_j - k_l - ic)$, with $E = \sum k_j^2$ ([Theorem 16.3](#), proved). The roots are real ([Proposition 16.4](#)), and each set of distinct I_j gives exactly one solution ([Proposition 16.5](#)).
- *One dimension*: two-body scattering can only permute momenta; three-body scattering can diffract, and non-diffraction is what the Bethe wavefunction asserts.
- *Labels*: $A' = Y(k_1 - k_2)A$, $Y_B = (kP - ic)/(k + ic)$, $Y_F = -(kP + ic)/(k + ic)$ ([Proposition 16.6](#)); in the momentum basis $X = PY$, $X_B = R(u)/(u + i)$ with $R(u) = u - iP$.
- *Yang–Baxter*: a consistent factorised three-body ansatz requires $X_{12}X_{13}X_{23} = X_{23}X_{13}X_{12}$ ([Theorem 16.8](#)); Yang’s matrix satisfies it because $u_{13} = u_{12} + u_{23}$ ([Proposition 16.9](#)); for a scalar S it is automatic. With unitarity it makes the N -body problem factorise (sketched).

Tutorial. Exercise Sheet 16.

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